Team 25 Team Charter

Team Members	1. Ragini Pawaiya(raginipawaiya5@gmail.com) 2. Shivanand Tiwari (0126CD231145@oriental.ac.in) 3. Kariuki Sophia (kariukisophia52@gmail.com) 4. Israel Georgewill(israelgeorgewill@gmail.com) 5. Esther Munyambu(esthermunyambu2004@gmail.co m) 6. chinomso Akumah(Essy4God@gmail.com) 7. Shivpratik Chaudhari(shivpratikc@gmail.com) …
Team Lead	Ragini Pawaiya (raginipawaiya5@gmail.com)
Team Members Roles and Responsibilities	Team Members Roles and Responsibilities Saint Louis University (SLU) — Office of International Services (OIS) Individual Company Contacts – Client, Role(s) in Company Ragini Pawaiya (raginipawaiya5@gmail.com) - Team Lead, represents team meetings, coordinates team activities, assigns tasks, and ensures project objectives are achieved. Kariuki Sophia (kariukisophia52@gmail.com) - Project Manager, provides guidance and support to team members, monitors project progress, and ensures that project activities remain on track. Shivanand Tiwari (0126cd231145@oriental.ac.in) - Project Scribe, responsible for taking meeting minutes, documenting discussions, maintaining project records, and assisting the Team Lead with communication. Israel Georgewill (israelgeorgewill@gmail.com) - Project Lead, responsible for monitoring task completion, tracking deadlines, and ensuring that project deliverables are completed on time. Esther Munyambu (esthermunyambu2004@gmail.com) - Data Quality Analyst, responsible for reviewing datasets, validating data accuracy, identifying data quality issues, and supporting data cleaning activities. Chinomso Akumah (Essy4God@gmail.com) - Research and Documentation Coordinator, responsible for gathering project-related information,

	supporting report preparation, maintaining documentation, and assisting with presentation materials. Shivpratik Chaudhari (shivpratikc@gmail.com) - Database and Dashboard Coordinator, responsible for supporting database management, data integration, dashboard development, and technical implementation tasks.
Mission, Vision Objectives & Core Values	*Mission :* To support Saint Louis University's Office of International Services by cleaning, integrating, and analysing international student data so that staff can confidently track I-901 fee payments, I-20 issuance, and deposit status without manual guesswork. *Vision Objectives :* Success to us means that by the end of this project, an OIS staff member can open a dashboard, see exactly which students are missing their SEVIS fee payment or still waiting on their I-20, and act on it the same day. We want our work to actually get used after we hand it over — not just sit in a report. As a team, we want to stay honest with each other about progress, divide work fairly, and not leave everything to one person at the last minute.
	Core Values: ***Integrity*** — we document what we actually did, including mistakes and limitations, as seen in our Week 1 report where we flagged every data gap honestly. ***Accountability*** — each member owns their deliverable for the week. If something is delayed, we say so early. ***Discipline*** — we stick to ISO date formats, clean naming conventions, and structured reporting because shortcuts create problems downstream. ***Respect*** — every team member's input gets heard in meetings, regardless of technical background. **Innovation** — we look for the smarter fix, like when we caught the snapshot duplication pattern in Connect instead of just accepting the inflated row count.

<table>
<tr>
<td>Internal Checks, Balances, and Reviews</td>
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Our team will work together to complete all project tasks on time and maintain the quality of our work. As the Team Leader, I will coordinate team activities, track progress, communicate with the sponsor, and ensure that deadlines are met.
We will hold a team meeting once every week to discuss completed work, review progress, solve any issues, and plan the next steps. Team members are expected to actively participate in meetings and share updates on their assigned tasks.
Before submitting any report or deliverable, the work will be checked by another team member to identify mistakes, missing information, or areas that need improvement. Necessary corrections will be made before the final submission.
SMART Goals

Specific: Complete all weekly project tasks and deliverables.
Measurable: Monitor progress through completed reports, data analysis tasks, and project milestones.
Achievable: Assign work according to each team member's skills and responsibilities.
Realistic: Set practical goals and deadlines that can be achieved within the project timeline.
Time-Based: Finish all assigned tasks before the scheduled submission dates.

Operations
Assignments: Tasks will be divided among team members based on project requirements and individual strengths.
Meetings: Weekly team meetings will be conducted to review progress and discuss upcoming work.
Communication Guidelines: Team members should respond to messages within 24 hours and inform the Team Leader if they face any challenges or delays.
Status Updates: Every team member will provide regular updates on their work during meetings.
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<table>
<tr><td></td><td>Deadlines: All tasks should be completed before the internal review date so that sufficient time is available for final checking and submission.</td></tr>
<tr><td>Operations:
Assignments
Meetings
Communication Guidelines
Status Updates
Deadlines</td><td> Assignments: Team members will work on data cleaning, data validation, database management, documentation, data analysis, and dashboard development according to their assigned responsibilities. Tasks will be distributed by the Team Leader based on project requirements and team members' skills. Meetings: The team will meet once every week through Microsoft Teams or Google Meet to discuss progress, review completed work, resolve issues, and plan upcoming tasks. Additional meetings will be scheduled when required. Communication Guidelines: The Team Leader will serve as the main point of contact with the sponsor. Team members are expected to communicate respectfully, actively participate in discussions, and respond to messages within 24 hours. WhatsApp and Microsoft Teams will be used for regular communication and updates. Status Updates: Each team member will provide weekly updates on completed tasks, current progress, and any challenges faced. Progress will be reviewed during team meetings before submission of weekly deliverables. Deadlines: All assigned tasks must be completed before the internal review date. Weekly reports and project deliverables will be submitted according to the timeline provided by Excelerate and the project sponsor. </td></tr>
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DATA PREPARATION REPORT

Week 1 — Dataset Overview, Cleaning, Integration & Validation

Prepared By	Team 25
Institution	Saint Louis University (SLU)
Submission Type	Week 1 Deliverable — Data Preparation
Datasets Covered	SEVIS Export, Applicant (Connect) Extract

Table of Contents

1. Executive Summary

This report documents the data preparation work completed during Week 1 of the international student compliance analytics project at Saint Louis University (SLU). The work covers three institutional datasets; the SEVIS DHS export, the Connect workflow extract, and a Google Sheets version of the Connect data, encompassing a combined raw record volume of over 72,000 rows prior to deduplication and cleaning.

3,524	6,875	72,389	160
SEVIS Students	Applicant Records	Raw Rows Processed	Columns Assessed

The SEVIS dataset contained 3,524 records of F-1 initial status students at SLU, exported from the DHS Student and Exchange Visitor Information System. After a thorough structural audit, 45 of 126 columns were removed due to being 100% null or holding a single constant value across all records. Multiple data corruption issues were identified and resolved, including Unix millisecond timestamps, Excel-encoded date values in numeric fields, and scientific notation ID formatting.

The Connect dataset, SLU's internal I-20 workflow tracking system, was found to be exported in a snapshot pattern, with the same 6,875 records duplicated five times in a single day, inflating the raw row count to 34,341. After deduplication, the dataset was confirmed at 6,875 unique applicant records. A Google Sheets version of the same data was identified as a duplicate source with degraded timestamp precision and was therefore excluded from the final data model.

A critical integration challenge was identified: no direct join key exists between the SEVIS and Applicant datasets. The SEVIS system uses SEVIS_ID as its primary identifier while Connect uses Reference_ID, these are from entirely different numbering systems with no overlap. Options for establishing the join are documented in Section 7.

Both cleaned datasets have been successfully uploaded to Supabase (PostgreSQL) and are ready for further analysis.

2. Objectives

This week's deliverable addresses the following analytical and operational objectives:

2.1 Primary Objectives

- Conduct a full structural audit of all three source datasets (SEVIS, Connect, Applicant).
- Identify and resolve data quality and validation issues including nulls, duplicates, data type mismatches, and formatting corruption.
- Standardize all variables; dates, categories, identifiers, and financial fields, to formats compatible with PostgreSQL and Supabase.
- Assess integration feasibility between datasets and document the join logic or its limitations.
- Produce clean, analysis-ready CSV files suitable for direct import into Supabase.

2.2 Secondary Objectives

- Document all transformation steps with before/after comparisons for institutional transparency.
- Establish a validated data foundation that Week 2 exploratory analysis and Week 3 dashboard development can reliably build upon.
- Identify structural compliance gaps, particularly around I-901 fee payment coverage, document submission completeness, and eligibility category distribution.
- Configure the Supabase database schema correctly, including primary keys, column types, and table relationships.

3. Introduction

3.1 Institutional Context

Saint Louis University (SLU) is a Jesuit research university located in St. Louis, Missouri. As a host institution under the Department of Homeland Security's Student and Exchange Visitor Program (SEVP), SLU is required to maintain accurate and up-to-date records for all F-1 and J-1 international students within the Student and Exchange Visitor Information System (SEVIS). The Office of International Services (OIS) is responsible for managing I-20 issuance, SEVIS record maintenance, and student compliance monitoring.

3.2 The Data Systems

This project works with data drawn from three interconnected institutional systems:

- SEVIS (Student and Exchange Visitor Information System): The federal DHS system that records all F-1 visa student data, including identity, academic programme, financial sponsorship, travel documents, and I-901 fee payment status. SLU exports this data periodically for institutional reporting and compliance monitoring.
- Connect: SLU's internal applicant workflow management system. Connect tracks the I-20 issuance pipeline, whether an applicant has paid their deposit, been assigned to an advisor, and had their I-20 issued. It bridges the admissions process with the SEVIS activation process.
- Applicant: A Google Sheets copy of the Connect data, used for ad-hoc access and sharing. This version was evaluated but found to be a duplicate of the Connect export with degraded data quality.

3.3 Project Motivation

The motivation for this project is to build a unified, validated analytical foundation that enables SLU's international services team to answer critical compliance questions: Which students have not yet paid their I-901 SEVIS fee? How many eligible students have had their I-20 issued versus those still pending? What is the deposit payment rate across the applicant pipeline? These questions cannot be answered reliably without first resolving the structural and quality issues identified in this report.

3.4 Scope & Limitations

This report covers data preparation only. Full exploratory data analysis (EDA), visualisation, and dashboard development are deferred to Weeks 2 and 3 respectively. The scope is limited to structural understanding, data quality rectification, and integration architecture. All personally identifiable information (PII) present in the datasets is handled in accordance with institutional data governance standards.

4. Methodology

4.1 Tools & Environment

Tool	Purpose	Version / Details
Excel	Data profiling, cleaning, and transformation	
Supabase	Cloud PostgreSQL database — data storage layer	Hosted PostgreSQL 15
pgAdmin 4	SQL interface for schema corrections and query development	pgAdmin 4
Looker Studio	Business intelligence and dashboard layer	Google Looker Studio
CSV Export	Data import mechanism into Supabase	Supabase CSV Importer (UI)

4.2 Process Overview

The data preparation process followed a structured five-stage pipeline:

- Stage 1 - Ingestion & Profiling: The raw files were loaded into excel. Row counts, column types, missing value rates, unique value distributions, and duplicate checks performed across all three datasets.
- Stage 2 - Structural Audit: Each column was evaluated for analytical utility. Columns were classified as: retain (analytically useful), drop-null (100% empty), or drop-constant (single value across all rows).
- Stage 3 - Cleaning & Transformation: The targeted corrections were applied; date format standardization, Unix timestamp decoding, Excel corruption removal, type casting, boolean conversion, and text normalization.
- Stage 4 - Deduplication. The snapshot-pattern duplicates in the Connect dataset identified and resolved by retaining only the latest record per Reference_ID.
- Stage 5 - Integration Assessment: Cross-dataset join keys evaluated. Integration constraints documented and resolution pathways proposed.

4.3 Quality Standards Applied

- ISO 8601 date format (YYYY-MM-DD) used for all date fields. This is the only format compatible with supabase.
- PostgreSQL-compatible Boolean (TRUE/FALSE) for all yes or no fields.
- Numeric types (NUMERIC/BIGINT/INTEGER) for all financial and ID fields.
- Uppercase standardization for all categorical variables.
- Email addresses normalized to lowercase.
- Column names free of special characters (parentheses, hyphens) for Supabase compatibility.

5. Dataset Overview & Documentation

5.1 SEVIS Dataset

5.1.1 Summary Statistics

3,524	126	81	45
Total Rows	Raw Columns	Cleaned Columns	Columns Dropped

5.1.2 Dataset Theme

For the SEVIS dataset, every record represents a single F-1 initial status international student, meaning the dataset captures the point of first I-20 issuance. It is the most comprehensive dataset in this project, covering student identity, academic programme details, financial sponsorship, travel documents, and government fee compliance.

5.1.3 Variable Classification

Variable Type	Count	Examples
ID Fields	7	SEVIS_ID, NonImmigrant_ID, FIN_ID, etc.
Date Fields	17	Date_of_Birth, Program_Start_Date, Program_End_Date.
Categorical	20	Gender, Education_Level, Country_of_Birth, Country_of_Citizenship.
Numerical (Financial)	8	Tuition_Fees, Living_Expenses, Dependent_Expenses, Other_Cost.
Boolean	3	No_Phone_Indicator, School_Require_English_Proficiency, Student_Has_English_Proficiency
Free Text / Address	21	Foreign_Address_1/2, Foreign_City, US_Physical_Address_1, Overall_Remarks, Preferred_Name
CIP Code (Text)	2	Major_1_Code, Major_2_Code

5.1.4 Key Identifiers

Field	Type	Role
SEVIS_ID	TEXT PK	Primary Key
NonImmigrant_ID	BIGINT	Secondary ID
FIN_ID	TEXT	Integration Key (sparse)
Email	TEXT	Cross-system link candidate

5.2 Applicant (Connect) Dataset

5.2.1 Summary Statistics

6,875	34,341	7	5×
Unique Applicants	Raw Rows (with dups)	Columns (cleaned)	Snapshot Copies

5.2.2 Dataset Theme

The Applicant dataset is an export from SLU's Connect workflow management system. It serves as the internal tracking layer for the I-20 issuance pipeline, recording whether each applicant has paid their enrollment deposit and whether their I-20 document has been issued. It is operationally focused, with minimal demographic data, primarily serving the administrative workflow of the Office of International Services.

5.2.3 Variable Classification

Column	Type	Description	Null %
Reference_ID	BIGINT (PK)	Primary identifier from Connect system	0%
Deposit_Status	BOOLEAN	Whether enrolment deposit has been paid	0%
I_20_Status	BOOLEAN	Whether I-20 document has been issued	0%
Assigned	TEXT	Advisor(s) assigned to this applicant	57.6%
Received_At	TIMESTAMP	Timestamp when record entered the system	0%
Created_At	TIMESTAMP	Record creation timestamp	0%
Modified_At	TIMESTAMP	Last modification timestamp	0%

5.2.4 Status Distribution

Status Field	TRUE (Yes)	FALSE (No)	% Completed
Deposit_Status	1,512	5,363	22.0%
I_20_Status	2,389	4,486	34.7%

Observation: More students have received their I-20 (2,389) than have paid their deposit (1,512). This appears counterintuitive as I-20 issuance typically llows deposit payment. This warrants investigation in Week 2 EDA.

5.3 Connect (Google Sheets) Dataset

VERDICT: EXCLUDED

This dataset is a duplicate of the Connect CSV export. It contains identical records but with degraded timestamp precision introduced by Google Sheets. It adds no unique value and should not be imported as a separate table.

Issue Details

- Same 6,875 unique Reference_IDs
- Same 5-snapshot duplicate pattern
- Created_At / Modified_At rounded to 2 values only

	◦ Timestamps lost granularity (e.g. 1,740,000,000,000 ms) ◦ Source CSVs (connect_cleaned.csv) preferred for all analysis

6. Data Cleaning & Processing Summary

6.1 SEVIS Cleaning

6.1.1 Before vs After — Structural

Dimension	Before (Raw)	After (Cleaned)	Action
Total Rows	3,524	3,524	No rows removed
Total Columns	126	81	45 columns dropped
100% Null Columns	14	0	All dropped
Single-Value Columns	31	0	All dropped
Date Format	MM/DD/YYYY (text)	YYYY-MM-DD (ISO 8601)	Standardised
Timestamp Format	Unix milliseconds (integer)	YYYY-MM-DD HH:MM:SS	Decoded
Financial Columns	String/text	NUMERIC	Type cast
Boolean Columns	Y / N strings	TRUE / FALSE	Converted
CIP Major Codes	Float (loses precision)	Text with 4 decimal places	Fixed
I_901_Transaction_Amount	Contains date value '12/15/1900'	Numeric only (corrupt → NULL)	Fixed
FIN_ID	Contains '#FMT' Excel errors	NULL where corrupt	Fixed
Number_of_Months_for_Funding	Contains date-encoded values	Correct integers	Fixed
Column name with parentheses	Number_of_Months_(for_funding)	Number_of_Months_for_Funding	Renamed
Text categoricals	Mixed case, whitespace	UPPERCASE, stripped	Standardised
Email	Mixed case	Lowercase, stripped	Standardised

6.1.2 Columns Dropped — 100% Null (14 columns)

These columns contained no data across all 3,524 records and were removed entirely:

Column Name	Reason for Null
Suffix	Not collected in this intake process
SEVIS_Legacy_First_Name	Legacy field from pre-2013 SEVIS system migration
SEVIS_Legacy_Middle_Name	Legacy field from pre-2013 SEVIS system migration
SEVIS_Legacy_Last_Name	Legacy field from pre-2013 SEVIS system migration
US_Mailing_Address_2	Students are pre-arrival; US addresses not yet established
US_Mailing_Routing_Code	Students are pre-arrival; US addresses not yet established

Termination_Reason	No terminations in this INITIAL status cohort
Other_Termination_Comment	No terminations in this INITIAL status cohort
Minor_Description	No dependant minor declarations made
Last_Session	Not applicable at INITIAL status stage
Study_Research_Abroad	Not applicable at INITIAL status stage
Thesis_Dissertation	Not applicable at INITIAL status stage
Cancellation_Reason	No cancellations in this extract
Agency_Code	No agency-sponsored students in this cohort

6.1.3 Columns Dropped — Single Constant Value (31 columns)

These columns held identical values for every one of the 3,524 records, providing zero analytical differentiation:

Column Name	Constant Value	Analytical Impact
SEVIS_Status	INITIAL	Entire export is INITIAL status — useless as a filter
Class_of_Admission	F-1	Entire export is F-1 class — useless as a filter
School_Name	Saint Louis University	Single-institution export — redundant
School_Code	STL214F00747000	Single-institution export — redundant
Campus_Name	Main Campus	Single campus — redundant
Campus_Code	STL214F00747000	Single campus — redundant
US_Mailing_Address_Status	ADDRESS_VALID_OVERRIDE	Uniform status — no variance
Other_Cost_Comment	Health Insurance	All students have same other cost type
Major_1_CIP_2020_Conversion	N	All records indicate no CIP conversion applied
Major_2_CIP_2020_Conversion	N	All records indicate no CIP conversion applied
Minor_CIP_2020_Conversion	N	No minors declared
Minor_Code	0	No minors declared
English_Is_Not_Required_Because	(single value)	Uniform across cohort
CPT_Approved	N	No CPT authorisations in this INITIAL cohort
OCE_Requested / OCE_Approved	N	No OCE activity in this cohort
Pre/Post_Completion_OPT (4 cols)	N	No OPT authorisations in this INITIAL cohort
STEM_OPT_Requested / Approved	N	No STEM OPT in this INITIAL cohort

Cap_Gap_Extended	N	No cap gap extensions in this cohort
Student request flags (5 cols)	N	No reinstatements, transfers, extensions requested
Border_Commuter_Student	N	No border commuter students
Correction_Request_Pending	N	No pending corrections
OPT_Portal	N	Not applicable at INITIAL status

6.1.4 Data Corruption Fixes — Detailed

Issue 1: Unix Millisecond Timestamps

Field	Raw Value (Example)	Corrected Value	Formula Used
Created_At	1744637845696	2025-04-14 13:37:25	divide by 1,000 → epoch seconds → datetime
Modified_At	1744637845696	2025-04-14 13:37:25	divide by 1,000 → epoch seconds → datetime

Formula: 1,744,637,845,696 ÷ 1,000 = 1,744,637,845.696 seconds since 1 Jan 1970 UTC = 2025-04-14 13:37:25. Without the ÷1000 correction, the value decodes to the year 57,000+.

Issue 2: Excel Cell Type Bleed — I_901_Transaction_Amount

Raw Values Found	Count	Correct Interpretation	Action
350	1,030	Valid — standard SEVIS fee	Retained
200	1	Valid — reduced fee category	Retained
12/15/1900	Unknown count	Invalid — Excel date in numeric field	Set to NULL
NULL	2,493	Student has not yet paid fee	Retained as NULL

Root cause: The source data was likely manipulated in Excel at some point. Excel auto-converted a numeric 0 or empty cell to a date value when the column type was changed, producing '12/15/1900' (day 15 of month 12 in Excel's 1900 date system).

Issue 3: Excel Cell Type Bleed — Number_of_Months_for_Funding

Raw Value	Count	Decoded Meaning	Corrected Value
9	1,806	9 months — already correct	9
01/08/1900	1,662	Month 8 encoded as Excel date	8
4	29	4 months — already correct	4
01/03/1900	19	Month 3 encoded as Excel date	3
01/11/1900	4	Month 11 encoded as Excel date	11
01/06/1900	2	Month 6 encoded as Excel date	6
12	1	12 months — already correct	12
01/04/1900	1	Month 4 encoded as Excel date	4

Root cause: Excel stores dates as integers counting from January 1, 1900. A month value of 8 was stored as the date January 8, 1900, then exported as '01/08/1900'. The day portion equals the original month number.

Issue 4: FIN_ID — Excel Format Error

Raw Value	Count	Cause	Action
#FMT	3 rows	Excel column was too narrow to display the number; wrote '#FMT' to the CSV instead	Replaced with NULL

6.1.5 Missing Value Analysis — SEVIS (columns > 5% missing)

Column	Missing Count	Missing %	Expected / Explanation
US Address fields (7 cols)	3,501–3,516	99.4–99.8%	Students are pre-arrival US addresses not yet established
US Telephone	3,521	99.9%	Pre-arrival no US number yet
Passport_Name	3,521	99.9%	Not required at INITIAL stage
City_of_Birth	3,490	99.0%	Optional field, rarely completed
Major_2_Code / Description	3,483	98.8%	Most students have one major only
FIN_ID	3,464	98.3%	Internal ID sparsely populated in SEVIS export
I_94 / Port of Entry / Visa fields	3,212–3,467	91–98%	Students have not yet entered the US
Foreign_State_Province	3,204	90.9%	Not all countries use state/province
I_901_Transaction_Amount	2,493	70.7%	53% haven't paid; 17% paid but amount not recorded
I_901_Transaction_Type / Receipt	1,869	53.0%	47% of students have not yet paid SEVIS fee
Financial columns (7 cols)	1,688–1,690	47.9%	Financial section not completed for ~half the cohort
Session dates (3 cols)	1,688	47.9%	Session scheduling not yet complete for new admits
Passport / Visa dates	1,628–1,685	46–48%	Documents not yet submitted or students pre-visa
I_901_Transaction_Date	1,064	30.2%	Payments made but date not recorded in ~30%
School_Fund_Type	239	6.8%	Minor gap — worth monitoring

6.2 Applicant (Connect) Cleaning

6.2.1 Before vs After — Structural

Dimension	Before (Raw)	After (Cleaned)	Action
Total Rows	34,341	6,875	Deduplicated (5× snapshot pattern removed)
Unique Reference_IDs	6,875	6,875	Confirmed unique after cleanup
Total Columns	26	7	19 columns dropped
100% Null Columns	15	0	All dropped
Single-Value Columns	2	0	Dropped (University, Total_Scholarship)
Ghost Columns	2	0	Unnamed: 24 and Unnamed: 25 dropped
Date Formats	Two different non-ISO formats	Unified YYYY-MM-DD HH:MM:SS	Standardised
Column Typo	Recieved_At	Received_At	Renamed
Hyphenated Column	I-20_Status	I_20_Status	Renamed (hyphen illegal in SQL)
Boolean Columns	Yes / No strings	TRUE / FALSE	Converted

6.2.2 The Snapshot Duplication Problem

The most significant issue in the Applicant dataset was not immediately obvious from a simple duplicate row check. The dataset contained 34,341 rows representing 6,875 unique students meaning each student appeared exactly 5 times. The cause was that the Connect system was exported 5 times on the same day (May 27, 2025) at different times, with each export capturing the full current state of all records.

Snapshot Timestamp	Row Count	Unique Reference_IDs	Duplicates Within Snapshot
4/14/2025 19:55	20	20	0
4/16/2025 16:06	1	1	0
5/27/2025 11:58	6,864	6,864	0
5/27/2025 12:00	6,864	6,864	0
5/27/2025 14:05	6,864	6,864	0
5/27/2025 15:02	6,864	6,864	0
5/27/2025 15:47	6,864	6,864	0
TOTAL	34,341	6,875	27,466 duplicates

Resolution: Records sorted by Received_At descending. Latest snapshot retained per Reference_ID using drop_duplicates(subset='Reference_ID', keep='first'). One record

(Reference_ID 1389787) had a genuine status change between snapshots (I_20_Status flipped from Yes to No) keeping the latest correctly captures the most current state.

6.2.3 Date Format Inconsistency

Column	Raw Format	Example	Cleaned Format
Received_At	MM/DD/YYYY HH:MM:SS	04/14/2025 19:55:52	2025-04-14 19:55:52
Created_At	M-D-YY H:MM AM/PM	4-14-25 2:25 PM	2025-04-14 14:25:00
Modified_At	M-D-YY H:MM AM/PM	5-27-25 10:17 AM	2025-05-27 10:17:00

Two completely different date formats existed within the same dataset. Received_At used a standard MM/DD/YYYY format while Created_At and Modified_At used a shorthand M-D-YY format with 12-hour AM/PM notation, a format typical of manual entry or a different system export configuration.

6.2.4 Missing Value Analysis — Applicant

Column	Missing Count	Missing %	Explanation
Assigned	3,962	57.6%	Advisors not yet assigned to majority of pipeline applicants
All other columns	0	0%	Fully populated

7. Data Integration & Validation

7.1 Current Database State (Supabase)

Table Name	Primary Key	Row Count	Status
Sevis	SEVIS_ID (TEXT)	3,524	Uploaded
Applicant	Reference_ID (BIGINT)	6,875	Uploaded

7.2 Integration Challenge — No Direct Join Key

A comprehensive analysis of both datasets confirmed that no direct join key exists between the Sevis and Applicant tables. The available options to resolve this is to include a Mapping Table. This is the cleanest solution. The Connect system almost certainly stores SEVIS_ID internally once the I-20 is issued. A single additional export column resolves the integration entirely.

7.3 Data Limitations & Risks

- No join key: Until a mapping table is obtained, cross-dataset analysis is not possible. Risk: compliance insights are isolated by the system.
- SEVIS financial data is 47.9% incomplete: Financial analysis covers only 1,836 of 3,524 students. Results may not represent the full cohort.
- Applicant Assigned field is 57.6% null: Advisor workload analysis will be unreliable for the majority of records.
- I_901 payment data is 53% null: Nearly half the cohort's fee payment status is unknown. Cannot confirm compliance for these students.
- Historical SEVIS records: Program_Start_Date ranges from 2000 to 2025. Some records may be legacy students rather than current intake. Cohort filtering may be needed in Week 2.

7.4 Compliance-Related Data Gaps

Gap Identified	Affected Records	Compliance Risk
No I_901 fee payment recorded	1,869 students (53%)	HIGH: cannot obtain visa without SEVIS fee receipt
No passport/visa information	3,200+ students (91%)	MEDIUM: expected pre-arrival, monitor as intake date approaches
No financial data on I-20	1,688 students (48%)	MEDIUM: I-20 cannot be issued without financial documentation
No advisor assigned (Applicant)	3,962 applicants (57.6%)	MEDIUM: no staff accountability for these cases
I-20 issued but no deposit paid	877 students*	LOW-MEDIUM: requires process verification
No join key between systems	All 3,524 + 6,875 records	HIGH: cannot produce unified compliance view

8. Conclusion

Week 1 has produced a cleaned, validated, and structurally documented foundation for the SLU international student compliance analytics project. The following outcomes have been achieved:

8.1 Achievements

- Three source datasets profiled, audited, and cleaned. SEVIS (3,524 records), Applicant/Connect (6,875 records after deduplication), and Connect Google Sheets (identified as duplicate, excluded)
- 45 columns removed from SEVIS and 19 from Applicant, all with documented rationale
- 27,466 duplicate rows removed from the Applicant dataset through snapshot deduplication
- Six distinct data corruption and format issues identified and resolved across both datasets
- All cleaned files successfully uploaded to Supabase as tables Sevis and Applicant
- Column type correction scripts written and partially executed in pgAdmin
- Integration architecture assessed, join key absence documented with four resolution pathways proposed
- Initial compliance gaps identified and quantified for Week 2 follow-up

8.2 Outstanding Items

Item	Priority	Owner	Target
Complete ALTER TABLE type corrections in pgAdmin (Sevis table)	HIGH	Data Team	Before Week 2
Resolve Number_of_Months_for_Funding UPDATE then ALTER	HIGH	Data Team	Before Week 2
Obtain mapping table or email field from Connect/OIS	HIGH	OIS / IT	Week 2
Verify I-20 without deposit anomaly (877 records)	MEDIUM	Data Team	Week 2 EDA
Filter historical SEVIS records to current intake cohort	MEDIUM	Data Team	Week 2 EDA
Connect Looker Studio to Supabase PostgreSQL endpoint	MEDIUM	Data Team	Week 2-3
Build compliance views in pgAdmin for Looker consumption	LOW	Data Team	Week 3

8.3 Readiness Assessment

95%	100%	60%	25%
Data Cleaning	DB Upload	Type Corrections	Integration

The data foundation is substantially ready for analytical use. The primary blocker for full readiness is the completion of Sevis~~vis~~ column type corrections in pgAdmin and the resolution of the cross-system join key. Once these are addressed, the dataset is positioned to support robust SEVIS compliance reporting, I-20 pipeline dashboards, and student eligibility analysis in Weeks 2 and 3.